PROBLEM BASED ASSESSMENT

Firewall Solution Proposal for Armadajaya Engineering Sdn Bhd

DFK50093 Computer Network Security | DIT5F | Session 1 2026/2027

Muhammad Haqimi bin Zailani (25DIT24F1110) & Syahir Iqwan bin Arfalmalus (25DIT24F1111)

Lecturer: Sir Mohd Fakhru Razie bin Mukhtar | Politeknik Malaysia Mersing

Why Firewall Deployment Matters

Firewalls are the primary perimeter defence: they inspect, filter and control traffic using predefined security rules.

1

INSIDE (LAN)

Trusted corporate workstations

2

DMZ

Semi-trusted public web services

3

OUTSIDE (WAN)

Untrusted internet and threats

Segmentation prevents unauthorised lateral movement and isolates critical internal assets.

Firewall Selection: SPI / NGFW

SPI / NGFW mechanism configured on an Edge Enterprise Router (Edge-Router-FW)

Stateful Session Tracking

Tracks TCP/UDP session state and automatically permits returning traffic for legitimate internal requests.

Granular Extended ACLs

Filters by source/destination IP, protocol and port (HTTP 80, HTTPS 443, SSH 22).

Zone-Based Isolation

Stops untrusted WAN traffic from initiating connections into the internal LAN.

Why not stateless? Stateless firewalls inspect packets in isolation. SPI/NGFW evaluates context in a session table, giving high throughput, preventing IP spoofing and cutting admin overhead.

Network Topology

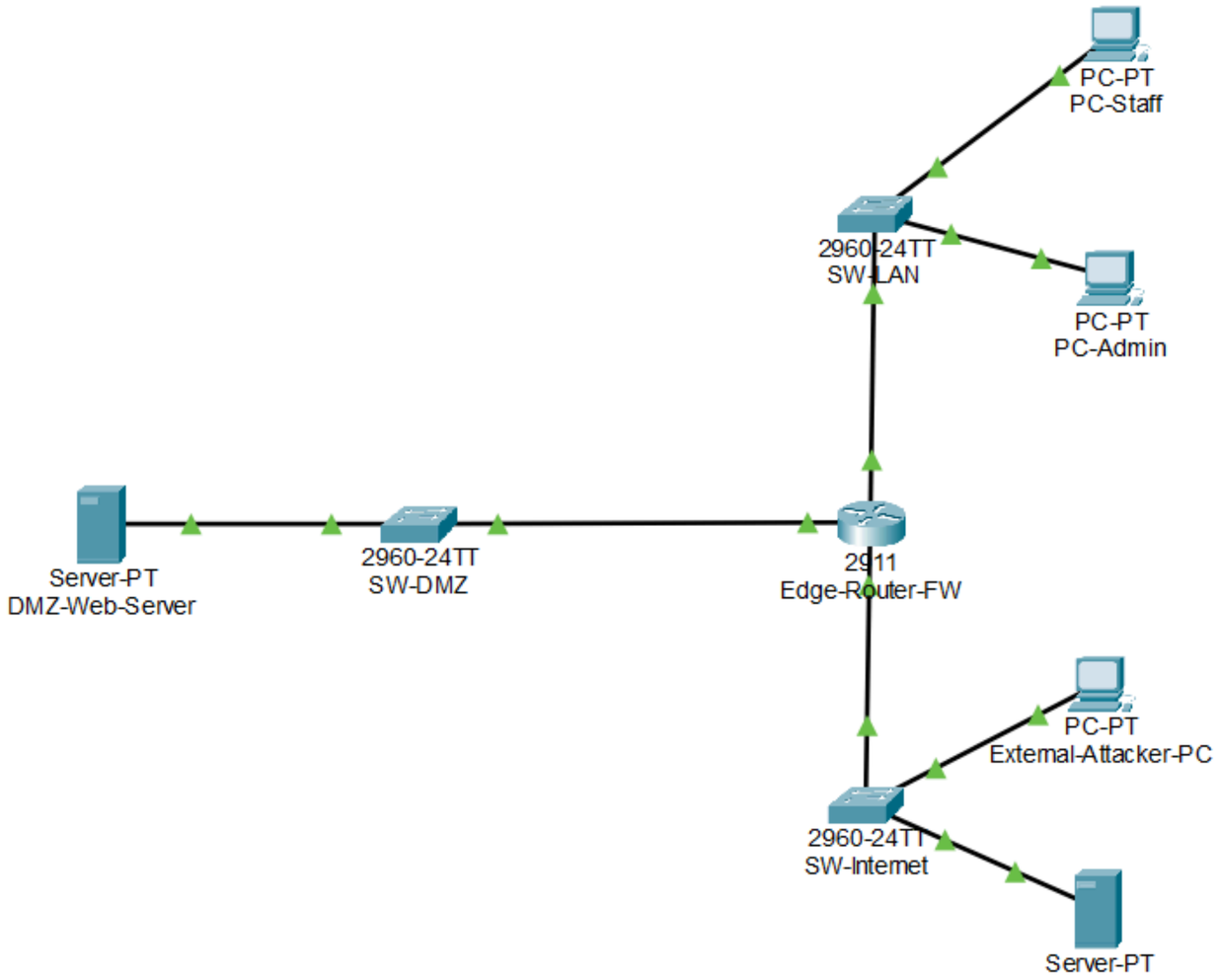


Figure 1: Packet Tracer topology, all zones meet at Edge-Router-FW

Inside Zone

PC-Admin and PC-Staff on the trusted LAN

DMZ Zone

DMZ-Web-Server hosts public services, separated from LAN

Outside Zone

External-Attacker-PC and Unsafe-Malicious-Server simulate the internet

IP Addressing Table

Zone	Device	Interface	IP Address	Gateway	Role
Inside	Edge-Router-FW	Gi0/0	172.16.10.1/24	N/A	LAN gateway
Inside	PC-Admin	Fa0	172.16.10.5/24	172.16.10.1	IT admin workstation
Inside	PC-Staff	Fa0	172.16.10.10/24	172.16.10.1	Employee device
DMZ	Edge-Router-FW	Gi0/1	172.16.20.1/24	N/A	DMZ gateway
DMZ	DMZ-Web-Server	Fa0	172.16.20.10/24	172.16.20.1	Public web server
Outside	Edge-Router-FW	Gi0/2	200.100.10.1/24	N/A	WAN edge gateway
Outside	External-Attacker-PC	Fa0	200.100.10.2/24	200.100.10.1	External client
Outside	Unsafe-Malicious-Server	Fa0	200.100.10.50/24	200.100.10.1	Simulated threat host

Initial Interface Configuration

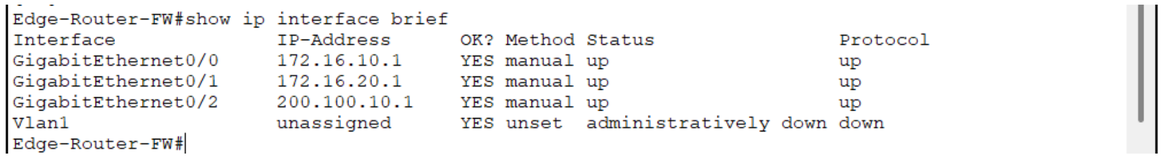
```
hostname Edge-Router-FW

! LAN
interface GigabitEthernet0/0
 ip address 172.16.10.1 255.255.255.0
 no shutdown

! DMZ
interface GigabitEthernet0/1
 ip address 172.16.20.1 255.255.255.0
 no shutdown

! WAN
interface GigabitEthernet0/2
 ip address 200.100.10.1 255.255.255.0
 no shutdown
```

Verification: show ip interface brief



```
Edge-Router-FW#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     172.16.10.1     YES manual up                    up
GigabitEthernet0/1     172.16.20.1     YES manual up                    up
GigabitEthernet0/2     200.100.10.1    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Edge-Router-FW#
```

All three interfaces are up/up with the correct static IPs, one per security zone.

Firewall Security Rules Matrix

Rule	Interface	Action	Source	Destination	Protocol / Port	Purpose
1	Gi0/0 IN	**DENY**	ANY	200.100.10.50	IP (all)	Block access to malicious host
2	Gi0/0 IN	**PERMIT**	172.16.10.5 (Admin)	ANY	TCP 22, 443	Admin-only management
3	Gi0/2 IN	**PERMIT**	ANY (Internet)	172.16.20.10	TCP 80, 443	Public web into DMZ
4	Gi0/0 IN	**PERMIT**	172.16.10.0/24	ANY	TCP 80, 443, UDP 53	Staff browsing and DNS
5	In / Out	**DENY**	ANY	172.16.10.0/24	IP (all)	Block internet from reaching LAN

Five primary policies enforced on the edge firewall.

Extended ACL Configuration

```
Edge-Router-FW(config)#access-list 101 deny ip any host 200.100.10.50
Edge-Router-FW(config)#access-list 101 permit tcp host 172.16.10.5 any eq 22
Edge-Router-FW(config)#access-list 101 permit tcp host 172.16.10.5 any eq 443
Edge-Router-FW(config)#access-list 101 permit tcp 172.16.10.0 0.0.0.255 any eq 80
Edge-Router-FW(config)#access-list 101 permit tcp 172.16.10.0 0.0.0.255 any eq 443
Edge-Router-FW(config)#access-list 101 permit tcp 172.16.10.0 0.0.0.255 any eq 53
Edge-Router-FW(config)#access-list 101 deny ip any any
Edge-Router-FW(config)#access-list 102 permit tcp any host 172.16.20.10 eq 80
Edge-Router-FW(config)#access-list 102 permit tcp any host 172.16.20.10 eq 443
Edge-Router-FW(config)#access-list 102 deny ip any 172.16.10.0 0.0.0.255
Edge-Router-FW(config)#access-list 102 permit ip any any
Edge-Router-FW(config)#interface GigabitEthernet0/0
Edge-Router-FW(config-if)#ip access-group 101 in
Edge-Router-FW(config-if)#exit
Edge-Router-FW(config)#interface GigabitEthernet0/2
Edge-Router-FW(config-if)#ip access-group 102 in
Edge-Router-FW(config-if)#exit
Edge-Router-FW(config)#end
Edge-Router-FW#
%SYS-5-CONFIG_I: Configured from console by console
write memory
```

```
Edge-Router-FW#show access-lists
Extended IP access list 101
    10 deny ip any host 200.100.10.50
    20 permit tcp host 172.16.10.5 any eq 22
    30 permit tcp host 172.16.10.5 any eq 443
    40 permit tcp 172.16.10.0 0.0.0.255 any eq www
    50 permit tcp 172.16.10.0 0.0.0.255 any eq 443
    60 permit tcp 172.16.10.0 0.0.0.255 any eq domain
    70 deny ip any any
Extended IP access list 102
    10 permit tcp any host 172.16.20.10 eq www (6 match(es))
    20 permit tcp any host 172.16.20.10 eq 443
    30 permit icmp any any
    40 deny ip any 172.16.10.0 0.0.0.255
    50 permit ip any any

Edge-Router-FW#
```

Figure 3: ACL commands and show access-lists output

ACL 101 on Gi0/0 (in)

Denies the malicious host, permits Admin SSH/HTTPS and LAN web/DNS, then denies the rest.

ACL 102 on Gi0/2 (in)

Permits HTTP/HTTPS to the DMZ server, allows ICMP, denies traffic to the LAN, permits the rest.

Test 1: Block Malicious Server (Rule 1)

Action

Ping from PC-Staff (172.16.10.10) to 200.100.10.50

Expected

Traffic blocked by Extended ACL 101

Actual

Request timed out, 4 sent, 0 received (100% loss)

PASSED

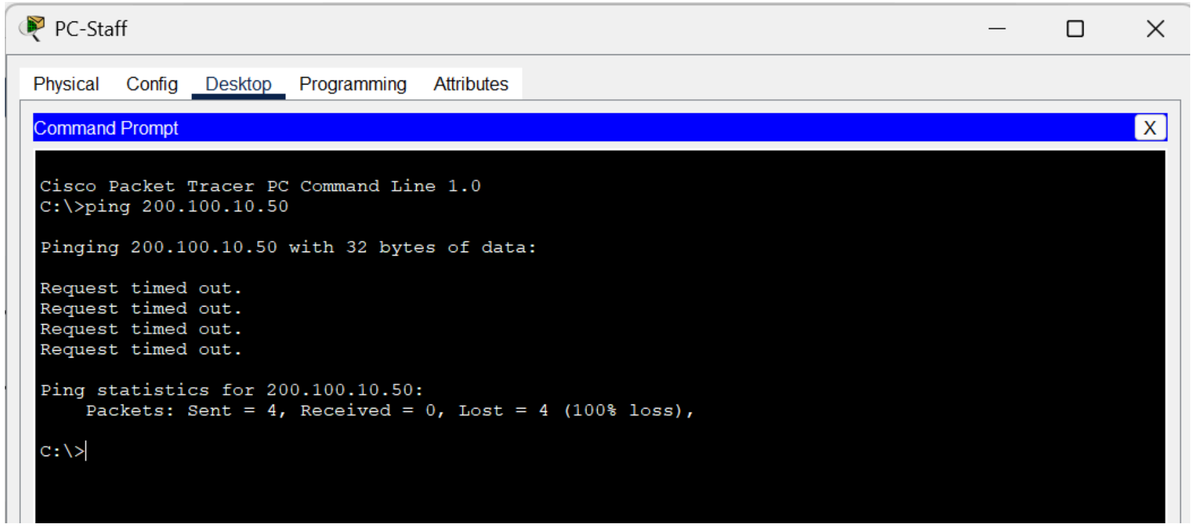


Figure 4: Ping to 200.100.10.50 blocked by ACL 101

Test 2: Web Access to DMZ Server (Rule 3)

Action

Browse http://172.16.20.10 from External-Attacker-PC (200.100.10.2)

Expected

HTTP response and webpage rendered from the DMZ server

Actual

Packet Tracer web server default page rendered

PASSED

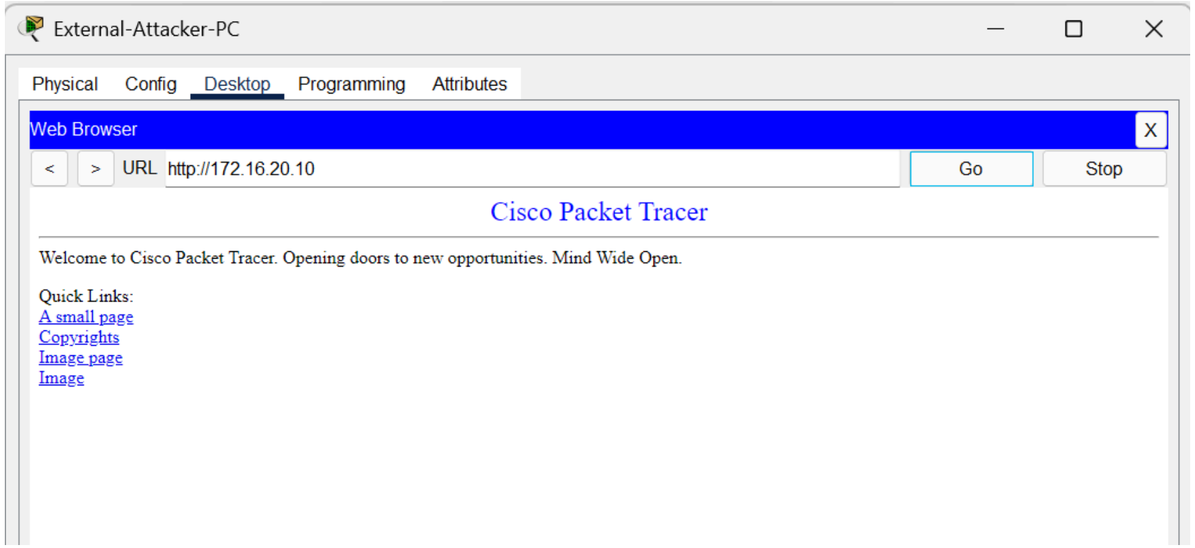


Figure 5: DMZ web page reached from the outside

Conclusion

Segmented

DMZ isolates public services while the LAN stays hidden from external exposure.

Protected

Malicious destinations are blocked from internal users.

Controlled

Admin access is limited to authorised endpoints; only legitimate public traffic reaches the DMZ.

Extended ACLs on Edge-Router-FW secure the enterprise topology, and both test cases passed.

THANK YOU